

CSA1717-ARTIFICIAL INTELLIGENCE

PRACTICAL LAB EXPERIMENTS

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PLANETS DATABASE

Aim:

To create a Prolog database containing information about planets and retrieve their properties.

Objective:

To represent planetary information using facts and retrieve planets based on their properties.

Algorithm:

- ☐ **Start.**
- ☐ **Define planet facts.**
- ☐ **Store properties such as type and order.**
- ☐ **Query the database.**
- ☐ **Display matching planets.**
- ☐ **Stop.**

Flowchart:

START

|

v

Create Planet Database

|

v

Enter Property/Planet

|

v

Search Database

|

v

Display Matching Planet

|

v

STOP

Prolog Program:

% Planets Database

planet(mercury, terrestrial, 1).

planet(venus, terrestrial, 2).

planet(earth, terrestrial, 3).

planet(mars, terrestrial, 4).

planet(jupiter, gas_giant, 5).

planet(saturn, gas_giant, 6).

planet(uranus, ice_giant, 7).

planet(neptune, ice_giant, 8).

terrestrial(P) :-

planet(P, terrestrial, _).

gas_giant(P) :-

planet(P, gas_giant, _).

ice_giant(P) :-

planet(P, ice_giant, _).

Output:

```
73 ?- consult(['c:/Users/Niharika Preejesh/OneDrive/Documents/planet1.pl']).  
true.  
  
74 ?- planet_type(earth, Type).  
Type = terrestrial.
```

Result:

Thus, the Prolog program successfully creates and queries a planets database.